Advanced MERN Stack Development
(Common to CSE, IT, AIML & CSE (DS))

	L	T	P	C
Course Code: 241CS018	**0**	**0**	**2**	**2**

Course Outcomes:
At the end of the course, student will be able to:

CO1: Make use of router, template engine and authentication using sessions to develop application in ExpressJS.
CO2: Build a single page application using RESTful APIs in ExpressJS.
CO3: Make use of components, props, stats and render data in ReactJS.
CO4: Apply router and hooks in designing ReactJS application.
CO5: Make use of MongoDB queries to perform CRUD operations on document database.

Mapping of Course Outcomes with Program Outcomes:

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11
CO1	2	2	3		2			1	2		2
CO2	2	3	2		2			1	2		2
CO3	2	2	3		2			1	2		2
CO4	2	3	2		2			1	2		2
CO5	2	2	3		2			1	2		2

Mapping of Course Outcomes with Program Specific Outcomes:

CO/PSO	PSO1	PSO2
CO1	2	
CO2	2	
CO3	2	
CO4	2	
CO5	2	

PRACTICE:

1. **ExpressJS – Routing, HTTP Methods, Middleware.**
   a. Write a program to define a route, Handling Routes, Route Parameters, Query Parameters and URL building.
   b. Write a program to accept data, retrieve data and delete a specified resource using http methods.
   c. Write a program to show the working of middleware.

2. **ExpressJS – Templating, Form Data**
   a. Write a program using templating engine.
   b. Write a program to work with form data.

3. ExpressJS – Cookies, Sessions, Authentication

a. Write a program for session management using cookies and sessions.
b. Write a program for user authentication.

4. ExpressJS – Database, RESTful APIs

a. Write a program to connect MongoDB database using Mangoose and perform CRUD operations.
b. Write a program to develop a single page application using RESTful APIs.

5. ReactJS – Render HTML, JSX, Components – function & Class

a. Write a program to render HTML to a web page.
b. Write a program for writing markup with JSX.
c. Write a program for creating and nesting components (function and class).

6. ReactJS – Props and States, Styles, Respond to Events

a. Write a program to work with props and states.
b. Write a program to add styles (CSS & Sass Styling) and display data.
c. Write a program for responding to events.

7. ReactJS – Conditional Rendering, Rendering Lists, React Forms

a. Write a program for conditional rendering.
b. Write a program for rendering lists.
c. Write a program for working with different form fields using react forms.

8. ReactJS – React Router, Updating the Screen

a. Write a program for routing to different pages using react router.
b. Write a program for updating the screen.

9. ReactJS – Hooks, Sharing data between Components

a. Write a program to understand the importance of using hooks.
b. Write a program for sharing data between components.

10. ReactJS Applications – To-do list and Quiz

a. Design to-do list application.

11. MongoDB – Installation, Configuration, CRUD operations

a. Install MongoDB and configure ATLAS
b. Write MongoDB queries to perform CRUD operations on document using insert(),

find(), update(), remove()

12. MongoDB – Databases, Collections and Records

a. Write MongoDB queries to Create and drop databases and collections.
b. Write MongoDB queries to work with records using find(), limit(), sort(), createIndex(), aggregate().

Additional Practice:

1. Design a to-do list application using NodeJS and ExpressJS.
2. Design a Quiz app using ReactJS.
3. Complete the MongoDB certification from MongoDB University website.

Note: The student must Complete & Submit a Express Js, React Js & MongoDB Certificate Courses offered by Infosys Spring board at the end of the Practice Session.

Reference Books:

1 Programming the World Wide Web, 7th Edition, Robet W Sebesta, Pearson.ISBN 13 9789332518827

2 Pro MERN Stack: Full Stack Web App Development with Mongo, Express, React, and Node, Vasan Subramanian, APress, O'Reilly, 2nd edition.ISBN 9781484226520.

Web Links:

1 https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_0138432901959516165 5073_shared/overview (Express Js)

2 https://infyspringboard.onwingspan.com/web/en/app/toc/lex_10648877150323546000_sha red/overview(ReactJs)

3 https://infyspringboard.onwingspan.com/en/app/toc/lex_auth_01317716929471283211 3_s hared/overview (MongoDB)